Lecture (5)

Nile University

Analytical Geometry & Calculus I

MTH111

Indeterminate forms and L'Hopital's Rule

Indeterminate forms and L'Hopital's Rule

Objective:

Use L'Hopital's Rule to solve problems involving limits.



Guillaume De l'Hôpital
1661 - 1704

Introduction Indeterminate forms

Consider: $\lim_{x\to 2}\dfrac{x^2-4}{x-2}$ or $\lim_{x\to 1}\dfrac{\ln x}{x-1}$

If we try to evaluate by direct substitution, we get: $\dfrac{0}{0}$

Zero divided by zero can not be evaluated. The limit may or may not exist, and is called an **indeterminate form**.

In the case of the first limit, we can evaluate it by factoring and canceling:

$$\lim_{x\to 2}\frac{x^2-4}{x-2} = \lim_{x\to 2}\frac{(x+2)(x-2)}{x-2} = \lim_{x\to 2}(x+2) = 4$$

This method does not work in the case of the second limit.

L'Hopital's Rule

Suppose f and g are differentiable

$$\lim_{x \to a} f(x) = 0 \quad \text{and} \quad \lim_{x \to a} g(x) = 0$$

(In other words, we have an indeterminate form of type $\frac{0}{0}$)

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)}$$

L'Hopital's Rule

$$\lim_{x\to a}\frac{f(x)}{g(x)} = \lim_{x\to 2}\frac{x^2-4}{x-2} = \frac{0}{0} = \lim_{x\to 2}\frac{\frac{d}{dx}\left(x^2-4\right)}{\frac{d}{dx}\left(x-2\right)} = \lim_{x\to 2}\frac{2x}{1} = 4$$

L'Hôpital's Rule:

If $\lim_{x\to a}\frac{f(x)}{g(x)}$ **is indeterminate** $\frac{0}{0}$ *OR* $\frac{\infty}{\infty}$ **, then:**

$$\lim_{x\to a}\frac{f(x)}{g(x)} = \lim_{x\to a}\frac{f'(x)}{g'(x)}$$

Proof.. later

L'Hopital's Rule

Example:

$$\lim_{x\to 1}\frac{\ln x}{x-1}=\frac{0}{0}$$

$$\lim_{x\to 1}\frac{\ln x}{x-1}=\lim_{x\to 1}\frac{\frac{d}{dx}(\ln x)}{\frac{d}{dx}(x-1)}=\lim_{x\to 1}\frac{1/x}{1}=\lim_{x\to 1}\frac{1}{x}=1$$

$$\lim_{x\to\infty}\frac{e^x}{x}=\frac{\infty}{\infty}=\lim_{x\to\infty}\frac{e^x}{1}=\infty$$

Remember:

$e^{\infty}=\infty \qquad e^{-\infty}=0$

$\ln 0=-\infty \qquad \ln\infty=\infty$

L'Hopital's Rule

Example: **Evaluate** $\lim_{x\to 0}\dfrac{1-\cos x}{x+x^2} = \dfrac{0}{0}$

$$\lim_{x\to 0}\frac{\sin x}{1+2x} = \frac{\sin(0)}{1+2(0)} = \frac{0}{1} = 0$$

L'Hopital's Rule

Example:

(i) $\lim_{x\to 0} \frac{x-\tan 3x}{x+\sin 5x}$ **(ii)** $\lim_{x\to 0} \frac{x^2-x\sin 8x}{x^3+4\tan^2 2x}$

(i) $$\lim_{x\to 0} \frac{x-\tan 3x}{x+\sin 5x} = \frac{0}{0}$$

$$= \lim_{x\to 0} \frac{1-3\sec^2 3x}{1+5\cos 5x}$$

$$= \frac{1-3}{1+5} = -\frac{1}{3}$$

(ii) (Try to solve)

L'Hopital's Rule

Example:

(i) $\lim_{x\to 1} \frac{3-3^x}{2-2^x}$ **(ii)** $\lim_{x\to 2} \frac{16-x^4}{16-4^x}$

(i) $\lim_{x\to 1} \frac{3-3^x}{2-2^x} = \frac{0}{0}$

$= \lim_{x\to 1} \frac{0-3^x \ln 3}{0-2^x \ln 2}$

$= \frac{3 \ln 3}{2 \ln 2}$

(ii) (Try to solve)

L'Hopital's Rule

Example: **Evaluate**

$$\lim_{x\to 0}\frac{\tan^{-1}x}{x}=\frac{0}{0}$$

$$\lim_{x\to 0}\frac{\left(\frac{1}{1+x^2}\right)}{1}=\frac{1}{1}=1$$

Remember:

$$\frac{d}{dx}(\tan^{-1}x)=\frac{1}{1+x^2}$$

L'Hopital's Rule

you can apply L'Hôpital's rule as many times as necessary as long as the fraction is still indeterminate:

EX: $$\lim_{x\to 0}\frac{e^x - x - 1}{x^2} = \frac{0}{0}$$

$$\lim_{x\to 0}\frac{e^x - 1}{2x} = \frac{0}{0}$$ ← Still $\frac{0}{0}$

$$\lim_{x\to 0}\frac{e^x}{2} = \frac{1}{2}$$ ← L'Hôpital again.

L'Hopital's Rule

Example: **Evaluate**

$$\lim_{x\to 0}\frac{\sinh x - x}{x^3} = \frac{0}{0}$$

$$\lim_{x\to 0}\frac{\cosh x - 1}{3x^2} = \frac{0}{0}$$

$$\lim_{x\to 0}\frac{\sinh x}{6x} = \frac{0}{0}$$

$$\lim_{x\to 0}\frac{\cosh x}{6} = \frac{1}{6}$$

L'Hopital's Rule

Example: (Try)

$$\lim_{x \to \infty} \frac{x^{10}}{e^x}$$

L'Hopital's Rule

Example: **Evaluate**

$$\lim_{x\to\infty}\frac{e^{2x}+x^2}{3e^{2x}+x^2} = \frac{\infty}{\infty}$$

$$= \lim_{x\to\infty}\frac{2e^{2x}+2x}{6e^{2x}+2x} = \frac{\infty}{\infty}$$

$$= \lim_{x\to\infty}\frac{4e^{2x}+2}{12e^{2x}+2} = \frac{\infty}{\infty}$$

$$= \lim_{x\to\infty}\frac{8e^{2x}}{24e^{2x}} = \lim_{x\to\infty}\frac{8}{24} = \frac{1}{3}$$

L'Hopital's Rule

Indeterminate forms

- $\frac{0}{0}$ or $\frac{\infty}{\infty}$
- $0 \times \infty$
- $\infty - \infty$
- $1^{\infty}, 0^{0}, \infty^{0}$

Case #1:- If $\lim_{x \to a} \frac{f(x)}{g(x)} = \frac{0}{0}$ or $\frac{\infty}{\infty}$, then

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)}$$

The others must be changed to fractions first.

L'Hopital's Rule(Indeterminate Products)

Example:

- *Evaluate the following limit.* $\lim_{x\to 0^+} x\ln x = 0\cdot -\infty$

Rewrite as a ratio!

$$\lim_{x\to 0^+}\frac{\ln x}{1/x}=\frac{-\infty}{+\infty}$$

Applying L'Hopital's Rule

$$=\lim_{x\to 0}\frac{\frac{1}{x}}{\frac{-1}{x^2}} \quad =\lim_{x\to 0}(-x) \quad =0$$

L'Hopital's Rule(Indeterminate Products)

Example: $\lim_{x\to\infty} xe^{-x} = \infty \cdot 0$

Rewrite as a ratio! $\lim_{x\to\infty} \frac{x}{e^x} = \frac{\infty}{\infty}$

$$\lim_{x\to\infty} \frac{1}{e^x} = \frac{1}{e^\infty} = 0$$

L'Hopital's Rule(Indeterminate Products)

Example:

$$\lim_{x\to\infty}\left(x\sin\frac{1}{x}\right) \qquad = \infty\cdot 0$$

Rewrite as a ratio!

$$\lim_{x\to\infty}\frac{\sin\frac{1}{x}}{\frac{1}{x}} \qquad = \frac{0}{0}$$

$$\lim_{x\to\infty}\frac{\sin\frac{1}{x}}{\frac{1}{x}} = \lim_{x\to\infty}\frac{\cos\left(\frac{1}{x}\right)\cdot\left(-\frac{1}{x^2}\right)}{-\frac{1}{x^2}} = \lim_{x\to\infty}\cos\left(\frac{1}{x}\right) = \cos(0) = 1$$

L'Hopital's Rule(Indeterminate Differences)

Example: $\lim_{x\to 0^+}\left(\frac{1}{x}-\frac{1}{\sin x}\right)$ ← This is indeterminate $\infty-\infty$

Rewrite as a ratio(by common denominator)

$\lim_{x\to 0}\left(\frac{\sin x - x}{x\sin x}\right)$ ⟵ Now it is in the form $\frac{0}{0}$

$$\lim_{x\to 0}\left(\frac{\cos x - 1}{x\cos x+\sin x}\right)=\frac{0}{0} \Rightarrow \lim_{x\to 0}\left(\frac{-\sin x}{-x\sin x+\cos x+\cos x}\right)$$

$$\lim_{x\to 0}\left(\frac{-\sin x}{-x\sin x+2\cos x}\right)=\frac{0}{2}=0$$

L'Hopital's Rule(Indeterminate Differences)

Example: $\lim_{x\to 1}\left(\frac{1}{\ln x}-\frac{1}{x-1}\right)$ ← This is indeterminate $\infty-\infty$

Rewrite as a ratio(by common denominator)

$$\lim_{x\to 1}\left(\frac{x-1-\ln x}{(x-1)\ln x}\right)$$ ← Now it is in the form $\frac{0}{0}$

$$\lim_{x\to 1}\left(\frac{1-\frac{1}{x}}{\frac{x-1}{x}+\ln x}\right)=\lim_{x\to 1}\left(\frac{x-1}{x-1+x\ln x}\right)=\frac{0}{0}$$

$$\lim_{x\to 1}\left(\frac{1}{1+1+\ln x}\right)=\frac{1}{2}$$

L'Hopital's Rule(Indeterminate powers)

Indeterminate powers: $1^{\infty}, 0^{0}, \infty^{0}$

Evaluating these forms requires a mathematical trick to change the expression into a ratio.

$$\ln u^{n} = n \ln u = \frac{\ln u}{\left(\frac{1}{n}\right)}$$

We can then write the expression as a ratio, which allows us to use L'Hôpital's rule.

L'Hopital's Rule(Indeterminate powers)

Example: $\lim_{x \to 0^+} x^x \leftarrow 0^0$

Let : $y = x^x$

Required : $\lim_{x \to 0} y = \lim_{x \to 0} x^x$

Take ln both sides: $\ln y = \ln x^x$

$\ln y = x \ln x$

Evaluate the new limit: $\lim_{x \to 0} \ln y = \lim_{x \to 0} x \ln x$

L'Hopital's Rule(Indeterminate powers)

Example: $\lim_{x\to 0} x^x$

Required : $\lim_{x\to 0} y = \lim_{x\to 0} x^x$

Evaluate the new limit: $\lim_{x\to 0} \ln y = \lim_{x\to 0} x \ln x$

$$\lim_{x\to 0} \frac{\ln x}{1/x} = \lim_{x\to 0} \frac{\frac{1}{x}}{\frac{-1}{x^2}} = \lim_{x\to 0}(-x) = 0$$

$$\therefore \quad \lim_{x\to 0} \ln y = 0 \Rightarrow \quad \lim_{x\to 0} y = e^0$$

$$\therefore \quad \lim_{x\to 0} x^x = e^0 = 1$$

L'Hopital's Rule(Indeterminate powers)

$$\lim_{x \to C} \left(f(x) \right)^{g(x)} = 1^{\infty}, \quad 0^{0} \text{ or } \quad \infty^{0}$$

Steps to solve Indeterminate powers:

1. Take ln for the given limit
2. Solve using **L'Hopital's**
3. Take exponential to the answer (from step 2)

L'Hopital's Rule(Indeterminate powers)

Example: $\lim_{x\to\infty}\left(1-\frac{3}{x}\right)^{x} = 1^{\infty}$

$$\lim_{x\to\infty}\ln(1-3/x)^{x} = \lim_{x\to\infty} x\ln(1-3/x) = \infty * 0$$

$$= \lim_{x\to\infty}\frac{\ln(1-3/x)}{1/x} = \frac{0}{0}$$

$$= \lim_{x\to\infty}\frac{\frac{-3(1/x)'}{1-3/x}}{(1/x)'} = \lim_{x\to\infty}\frac{-3}{1-3/x} = \mathbf{-3}$$

$$\therefore \quad \lim_{x\to\infty}(1-3/x)^{x} = e^{\mathbf{-3}}$$

L'Hopital's Rule(Indeterminate powers)

Example: $\lim_{x\to\infty}(e^x+1)^{\frac{1}{x}} \quad = \infty^0$

$$\lim_{x\to\infty}\ln(e^x+1)^{\frac{1}{x}} = \lim_{x\to\infty}\frac{1}{x}\ln(e^x-1)$$

$$= \lim_{x\to\infty}\frac{\ln(e^x+1)}{x} = \frac{\infty}{\infty}$$

$$= \lim_{x\to\infty}\frac{\frac{e^x}{e^x+1}}{1} = \lim_{x\to\infty}\frac{e^x}{e^x+1} = \frac{\infty}{\infty}$$

$$= \lim_{x\to\infty}\frac{e^x}{e^x} = 1$$

$$\therefore \quad \lim_{x\to\infty}(e^x+1)^{\frac{1}{x}} = e^1 = e$$

L'Hopital's Rule(Indeterminate powers)

Example: $\lim_{x\to 0^+}(e^x-1)^x = 0^0$

$$\lim_{x\to 0^+}\ln(e^x-1)^x = \lim_{x\to 0^+} x\ln(e^x-1) = 0*(-\infty)$$

$$= \lim_{x\to 0^+}\frac{\ln(e^x-1)}{1/x} = \frac{-\infty}{\infty}$$

$$= \lim_{x\to 0^+}\frac{\frac{e^x}{e^x-1}}{-\frac{1}{x^2}} = \lim_{x\to 0^+}\frac{-x^2e^x}{e^x-1} = \frac{0}{0}$$

$$= \lim_{x\to 0^+}\frac{-(2xe^x+x^2e^x)}{e^x} = \lim_{x\to 0^+}(-2x-x^2) = 0$$

$$\therefore \quad \lim_{x\to 0^+}(e^x-1)^x = e^0 = 1$$

Proof L'Hopital's Rule

Note:

$$f'(x)=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$$

At x=a

$$f'(a)=\lim_{h\to 0}\frac{f(a+h)-f(a)}{h}$$

Let y=a+h

$$f'(a)=\lim_{y\to a}\frac{f(y)-f(a)}{y-a}$$

Proof L'Hopital's Rule

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a} \qquad g'(a) = \lim_{x \to a} \frac{g(x) - g(a)}{x - a}$$

L'Hopital's Rule requires: $\lim_{x \to a} \frac{f(x)}{g(x)} = \frac{\lim_{x \to a} f(x)}{\lim_{x \to a} g(x)} = \frac{0}{0}$

therefore $f(a)\ \&\ g(a) \Rightarrow 0$

Working backwards:

$$\lim_{x \to a} \frac{f'(x)}{g'(x)} = \frac{\lim_{x \to a} \dfrac{f(x) - f(a)}{x - a}}{\lim_{x \to a} \dfrac{g(x) - g(a)}{x - a}} = \lim_{x \to a} \frac{\dfrac{f(x) - f(a)}{x - a}}{\dfrac{g(x) - g(a)}{x - a}}$$

$$= \lim_{x \to a} \frac{f(x) - f(a)}{g(x) - g(a)} = \lim_{x \to a} \frac{f(x) - 0}{g(x) - 0} = \lim_{x \to a} \frac{f(x)}{g(x)}$$



See you next week

L'Hopital's Rule

Example:

(i) $\lim_{x\to 4} \frac{\sin\left(\sqrt{x^2-7}-3\right)}{x^3-5x^2+3x+4}$ (ii) $\lim_{x\to 2} \frac{x^4-x^3-8}{\sin(x^3-x-6)}$

(i) $\lim_{x\to 4} \frac{\sin\left(\sqrt{x^2-7}-3\right)}{x^3-5x^2+3x+4} = \frac{0}{0}$

$$= \lim_{x\to 4} \frac{\left[\cos\left(\sqrt{x^2-7}-3\right)\right] * \frac{2x}{2\sqrt{x^2-7}}}{3x^2-10x+3}$$

$$= \frac{1 * \frac{4}{3}}{48-40+3} = \frac{4}{11*3} = \frac{4}{33}$$

(ii) (Try to solve)

L'Hopital's Rule

Example: **Evaluate**

$$\lim_{x\to\infty}\frac{1+\ln x}{2x+1} = \frac{\infty}{\infty}$$

$$= \lim_{x\to\infty}\frac{\frac{1}{x}}{2} = \frac{0}{2} = 0$$

L'Hopital's Rule

Example:

(i) $\lim_{x\to\left(\frac{\pi}{2}\right)^-} \frac{\ln(\cot x)}{\tan x}$ (ii) $\lim_{x\to 0^+} \frac{\ln(\tan 2x)}{\ln(\tan 5x)}$

(i) $\lim_{x\to\left(\frac{\pi}{2}\right)^-} \frac{\ln(\cot x)}{\tan x} = \frac{-\infty}{+\infty}$

$$= \lim_{x\to\left(\frac{\pi}{2}\right)^-} \frac{\frac{-\csc^2 x}{\cot x}}{\sec^2 x} = \lim_{x\to\left(\frac{\pi}{2}\right)^-} \frac{-\sin x \cos^2 x}{\sin^2 x \cos x}$$

$$= \lim_{x\to\left(\frac{\pi}{2}\right)^-} (-\cot x) = 0$$

(ii) (Try to solve)

L'Hopital's Rule(Indeterminate Products)

Example: $\lim_{x\to\pi/4}(1-\tan x)\sec 2x \quad = 0\cdot\infty$

$$\lim_{x\to\pi/4}\frac{1-\tan x}{\cos 2x}$$

$$\lim_{x\to\pi/4}\frac{-\sec^2 x}{-2\sin 2x}=\frac{-2}{-2}=1$$

L'Hopital's Rule(Indeterminate powers)

Example: $\lim_{x\to\infty}(1+e^x)^{\operatorname{sech} x} = \infty^0$

$$\lim_{x\to\infty} \ln(1+e^x)^{\operatorname{sech} x}$$

$$= \lim_{x\to\infty} (\operatorname{sech} x)\ln(1+e^x) = 0 * \infty$$

$$= \lim_{x\to\infty} \frac{\ln(1+e^x)}{\cosh x} = \frac{\infty}{\infty}$$

$$= \lim_{x\to\infty} \frac{\frac{e^x}{1+e^x}}{\sinh x} = \lim_{x\to\infty} \frac{\frac{1}{e^{-x}+1}}{\sinh x} = \frac{1}{\infty} = 0$$

$$\therefore \quad \lim_{x\to\infty}(1+e^x)^{\operatorname{sech} x} = e^0 = 1$$